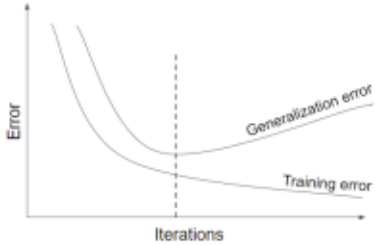
EARLY STOPPING

- Especially for complex nonlinear models we can easily overfit
- In optimization: Often, after a certain number of iterations, generalization error begins to increase even though training error continues to decrease



EARLY STOPPING AND $L2$

Goodfellow, Bengio, and Courville 2016

For iterative optimizers like SGD, we can monitor this step by step over small iterations:

1. Split train data $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{subtrain}}$ and $\mathcal{D}_{\text{val}}$ (e.g. with ratio of 2:1)
2. Train on $\mathcal{D}_{\text{subtrain}}$ and eval model on $\mathcal{D}_{\text{val}}$
3. Stop when validation error stops improving (with a margin of patience steps)
4. Use parameters of the previous step for the actual model

Strengths	Weaknesses
Effective and simple	Periodical evaluation of validation error
Applicable to almost any model without adjustment	Temporary copy of $\boldsymbol{\theta}$ (we have to save the whole model each time validation error improves)
Combinable with other regularization methods	Less data for training $\rightarrow$ include $\mathcal{D}_{\text{val}}$ afterwards

- For simple case of LM with squared loss and GD optim initialized at $\boldsymbol{\theta} = 0$: Early stopping has exact correspondence with $L2$ regularization/WD: optimal early-stopping iter T_{stop} inversely proportional to λ scaled by step-size α

More sophisticated forms also apply cross-validation.

$$T_{\text{stop}} \approx \frac{1}{\alpha\lambda} \Leftrightarrow \lambda \approx \frac{1}{T_{\text{stop}}\alpha}$$

- Small λ (regu. $\downarrow$) $\Rightarrow$ large T_{stop} (complexity $\uparrow$) and vice versa

SGD TRAJECTORY AND $L2$

Ali, Dobriban, and Tibshirani 2020

Solution paths for $L2$ regularized linear model closely matches SGD trajectory of unregularized LM initialized at $\theta = 0$

Strengths	Weaknesses
Effective and simple	Periodical evaluation of validation error
Applicable to almost any model without adjustment	Temporary copy of θ (we have to save the whole model each time validation error improves)
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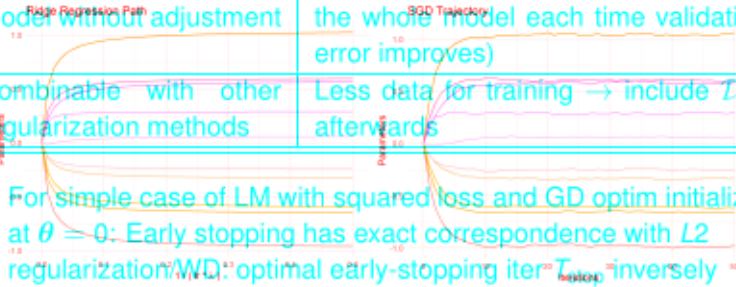

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Caveat: Initialization at the origin is crucial for this equivalence to hold, which is almost never exactly used in practice in ML/DL applications

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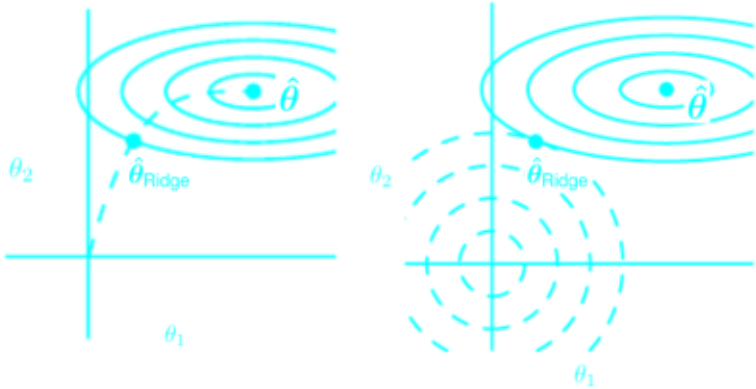
EARLY STOPPING AND $L2$ ▸ Goodfellow, Bengio, and Courville 2016 / 2

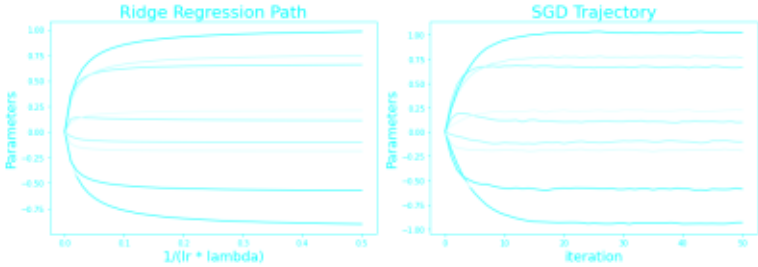


Goodfellow et al. (2016)

- Solid lines are $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
- LHS: Trajectory of GD early stopped, initialized at origin
- RHS: Constrained form of ridge regularization

SGD TRAJECTORY AND $L2$ ▸ Ali, Dobriban, and Tibshirani 2020

Solution paths for $L2$ regularized linear model closely matches SGD trajectory of unregularized LM initialized at $\boldsymbol{\theta} = 0$



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